

# Multi-View 3D Reconstruction Using Attention and Visual Odometry

Kamleshwar T, Mitran R, Sri Harsha Arigapudi, Vignesh N, Nitin S

*Department of Computer Science and Engineering, PSG College of Technology, Coimbatore, India*

## Abstract

Recovering a three-dimensional representation of a scene from ordinary RGB video, without relying on active depth sensors such as LiDAR, radar, or structured-light devices, remains an open problem in computer vision and robotics. Sensor-based systems are accurate but costly, difficult to maintain, and fragile to individual sensor failure. This work presents a multi-view 3D reconstruction pipeline that operates purely on RGB video and combines geometric visual odometry with a learned, attention-based multi-view feature-fusion network. A keyframe-selection stage retains frames whose relative motion exceeds a rotation/translation threshold; a hybrid geometric-and-deep-learning visual odometry module recovers each keyframe's camera pose; a lightweight MNASNet backbone with a Feature Pyramid Network (FPN) extracts multi-scale 2D features; and a geometric unprojection stage lifts these features into a sparse 3D voxel volume. The central contribution is an attention mechanism that replaces naive additive multi-view feature accumulation with learned, query-key weighted aggregation, allowing each voxel to weight its contributing views by relevance rather than treating every observation identically. A residual U-Net regresses a truncated signed distance function (TSDF) over the refined volume, from which a surface mesh is extracted via marching cubes. Trained and evaluated on the ScanNet indoor-scene dataset (6,000 training / 1,110 validation / 630 test scenes), the attention-augmented architecture reduces the L1 TSDF regression loss from 1.23 to 0.93 relative to a non-attention baseline. For visual odometry, a combined geometric-and-deep-learning estimator achieves an end-point error of 0.0919, outperforming geometry alone without scale correction (0.2151) and geometry with ground-truth scale correction (0.1478), despite operating without access to ground-truth scale. Reconstructed meshes closely match LiDAR-derived ground truth across evaluated test scenes. The primary limitation is computational cost: dense 3D convolutions and the absence of real-time scale recovery for monocular odometry currently preclude real-time deployment. We discuss this limitation together with directions for future work, including extending attention-based fusion to 3D semantic segmentation and a native C++ implementation of the attention module.

*Keywords: 3D reconstruction, multi-view vision, visual odometry, attention, TSDF, ScanNet, MNASNet, FPN, sparse 3D processing, deep learning*

## 1. Introduction

### Motivation

3D scene reconstruction from multi-view RGB images underpins autonomous navigation, SLAM, virtual and augmented reality, 3D printing, and medical and industrial imaging. Conventional autonomous systems reconstruct scenes by fusing data from cameras, LiDAR, radar, IMUs, time-of-flight sensors, and GPS. This sensor suite improves robustness but is costly to acquire and maintain, and makes the system fragile: failure of any single sensor can degrade the entire pipeline. This motivates a camera-centric alternative that substitutes deep learning and geometric algorithms for the missing active-sensor modalities, operating on RGB video alone.

## Problem Statement

Given a sequence of RGB observations of a scene, together with camera intrinsic parameters, the system must: (i) estimate the relative camera motion between frames; (ii) select a reduced set of informative keyframes; (iii) transform per-frame image features into a common 3D coordinate system; (iv) fuse multi-view feature observations at each 3D location; and (v) predict a volumetric representation of the underlying scene surface from the fused features.

## Objectives and Scope

The objective is to develop a deep model that regresses a signed distance value for each location in a 3D scene volume from RGB images together with camera extrinsic and intrinsic parameters, and to determine whether an attention mechanism improves multi-view feature fusion relative to naive additive fusion. The resulting pipeline targets autonomous systems, SLAM, and other settings requiring geospatial scene understanding from ordinary RGB cameras rather than dedicated depth sensors.

## Contributions

This work makes the following contributions:

1. A multi-stage RGB-only 3D reconstruction pipeline requiring no active depth sensor at inference time.
2. A hybrid visual-odometry module combining classical epipolar geometry with a learned, FlowNet-and-LSTM-based refinement stage.
3. Lightweight multi-scale 2D feature extraction using an ImageNet-pretrained MNASNet backbone with a Feature Pyramid Network.
4. A geometric 2D-to-3D unprojection stage that maps image features into a sparse voxel volume using per-view projection matrices.
5. An attention-based multi-view feature aggregation mechanism that replaces additive voxel-feature accumulation with learned, query-key weighted fusion.
6. Quantitative evaluation of both the reconstruction network (TSDF regression loss, occupancy confusion matrices) and the visual odometry module (end-point error) on the ScanNet dataset.

## 2. Background and Related Work

3D scene reconstruction has followed a fairly linear historical progression: from active LiDAR scanning, through classical epipolar-geometry reconstruction and learned fundamental-matrix estimation, to single- and then multi-view learned reconstruction — of which the attention-weighted multi-view fusion proposed in this work represents the latest step (Fig. 2).

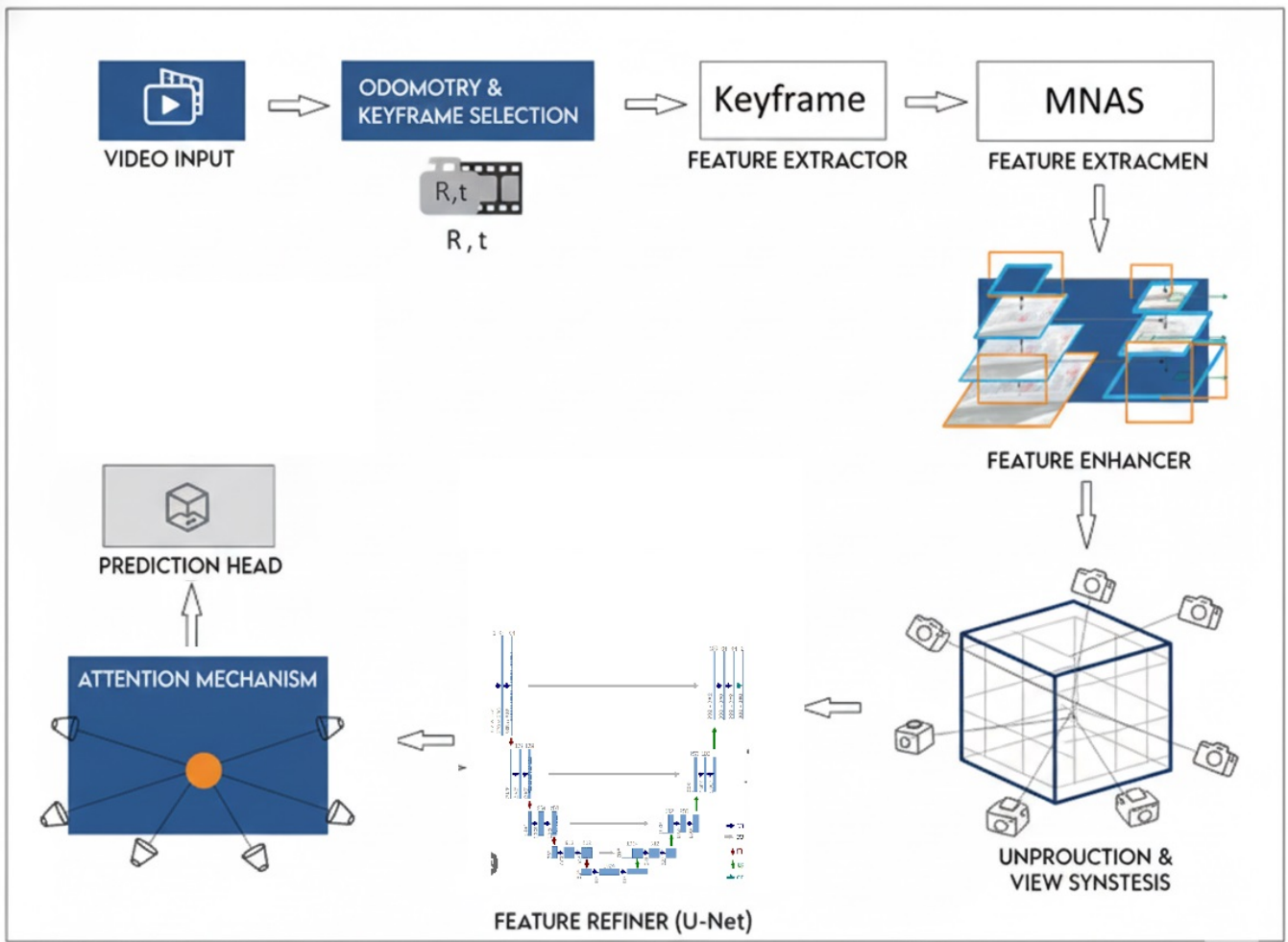


Figure 1: Overall system pipeline: video capture, keyframe selection, visual odometry, 2D feature extraction/enhancement, geometric unprojection, attention-based aggregation, and TSDF prediction.

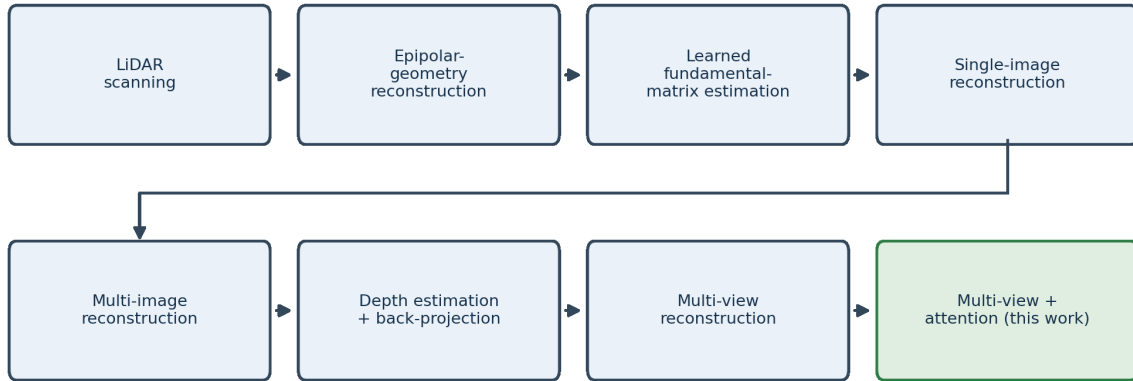


Figure 2: Evolution of 3D reconstruction approaches, from active sensing to the attention-augmented multi-view approach proposed in this work.

## Classical Feature Extraction and Multi-View Geometry

Scale-invariant feature extraction (SIFT) provides local image descriptors that remain matchable across changes in scale, rotation, and viewpoint, by locating extrema in a difference-of-Gaussian scale-space and describing each keypoint from its local gradient histogram [1]. Such correspondences are classically used to estimate the

fundamental or essential matrix relating two camera views via the epipolar constraint  $\mathbf{x}'^T \mathbf{E} \mathbf{x} = 0$ , from which relative camera rotation and translation can be recovered and 3D points triangulated [2]. In this project, classical feature-correspondence concepts provide background for the geometric component of the visual-odometry module described in Section 8, rather than constituting the primary learned reconstruction pathway; the implemented pipeline instead relies on optical-flow-based tracking and the essential matrix for its geometric pose estimate.

For completeness, the classical SIFT pipeline itself proceeds through scale-space extrema detection via a difference-of-Gaussian (DoG) pyramid (Fig. 3), sub-pixel keypoint localization via a Taylor expansion, elimination of edge-like responses using the principal-curvature ratio of the Hessian, orientation assignment from local gradient histograms, construction of a distinctive keypoint descriptor, and finally RANSAC-based correspondence filtering (Fig. 4). None of these classical steps are part of this project's learned pipeline; they are summarized here only as background for the geometric literature the odometry module draws on.

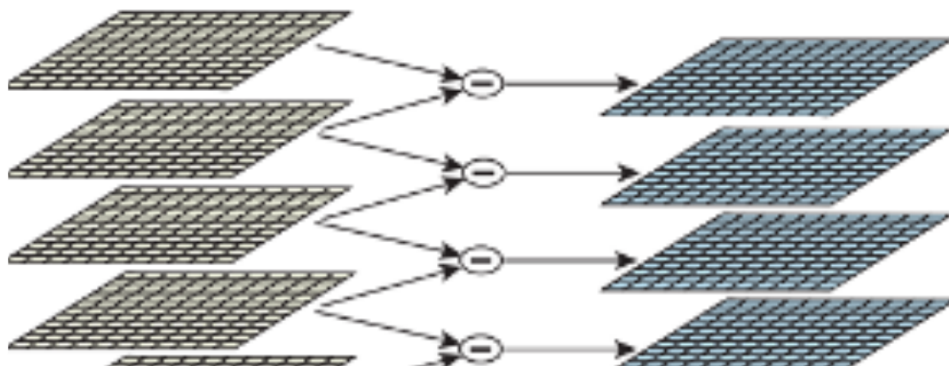


Figure 3: Difference-of-Gaussian (DoG) scale-space construction used by SIFT to detect scale-invariant extrema (classical background, not part of the proposed pipeline).

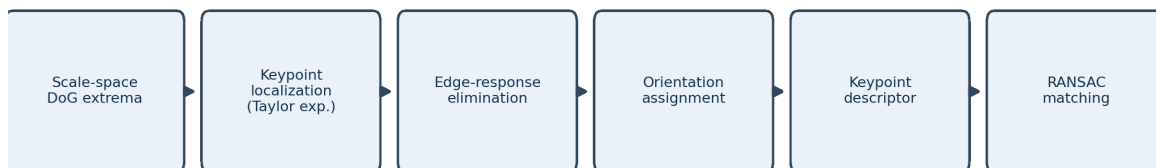


Figure 4: Classical SIFT feature pipeline, summarized at a high level.

Recovering relative camera pose from such correspondences classically proceeds by estimating the fundamental matrix  $\mathbf{F}$  from the epipolar constraint, decomposing it (together with an assumed skew form) into camera matrices  $\mathbf{P}$ ,  $\mathbf{P}'$ , and triangulating 3D points via SVD (Fig. 5, Fig. 6). As with the SIFT pipeline above, this fundamental-matrix route is background rather than a contribution of this work: the implemented visual-odometry module (Section 8) instead recovers pose via the essential matrix, using known camera intrinsics and Nister's five-point algorithm, which is more accurate than the uncalibrated fundamental-matrix route summarized here.

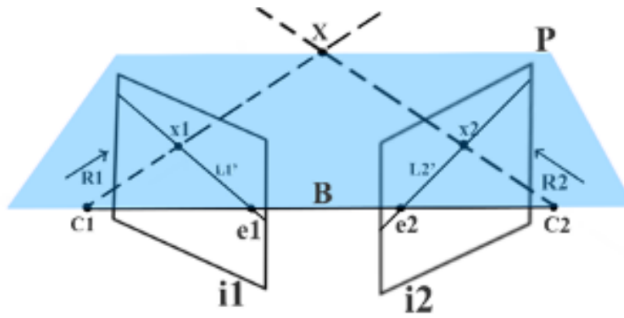


Figure 5: Epipolar geometry between two camera views, underlying classical fundamental-matrix pose recovery (classical background).

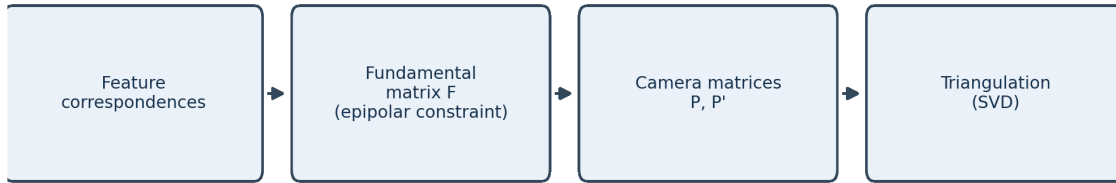


Figure 6: Classical pipeline for recovering camera pose and 3D structure from the fundamental matrix (background only; the implemented odometry module uses the essential matrix instead).

## Learned Visual Odometry

Wang et al. (DeepVO) estimate camera ego-motion end-to-end by feeding FlowNet-derived optical-flow features into a recurrent network that exploits temporal context across a frame sequence [3]. This project's visual-odometry module adopts the same general idea — combining an optical-flow-derived feature with a sequence model — but uses it to refine, rather than replace, a classical geometric pose estimate (Section 8).

## Learned 3D Reconstruction

Occupancy Networks represent a 3D surface as a continuous implicit function and predict occupancy probability for a query point conditioned on image features [4]. Among the family of 3D scene representations used across this literature — depth maps, voxel grids, point clouds, meshes, occupancy grids, signed distance functions, and truncated signed distance functions (TSDF) — this work adopts the TSDF representation, which combines an occupancy signal with sub-voxel surface localization and is well suited to fusion from noisy multi-view observations (formalized in Section 3). DISN extends implicit surface prediction to signed distance regression for single-view reconstruction using intermediate depth and normal maps [7]. SurfaceNet unprojects paired-view image and camera data into a shared 3D volume for occupancy prediction but is limited to two views [11]; NeuralRecon extends this to a multi-view setting with GRU-based fusion across frames [12]. DeepVideoMVS replaces expensive 3D convolutions with a lightweight convolutional-LSTM formulation to reduce inference cost [13]. Attention-based aggregation has separately been shown to improve multi-view fusion: Salvi et al. apply channel-wise self-attention within an occupancy-network encoder [8], and Yang et al. learn attention scores over a set of per-view latent features in place of sequential fusion [10]. This project's central contribution builds on this last line of work, applying learned attention directly to the voxel-level multi-view accumulation step of a SurfaceNet/NeuralRecon-style pipeline.

## Comparison of Related Approaches

| Approach                            | Main Idea                                                          | Relevance to This Work                                                    |
|-------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------|
| SIFT / classical features [1, 2]    | Local invariant keypoint descriptors and epipolar geometry         | Background for geometric correspondence in the odometry module            |
| DeepVO [3]                          | Learned visual odometry via FlowNet + RNN                          | Basis for the learned-refinement half of the hybrid odometry module       |
| Occupancy Networks [4]              | Implicit continuous occupancy function                             | Related representation; this work uses TSDF instead                       |
| SurfaceNet [11]                     | Two-view unprojection + 3D refinement for occupancy                | Base unprojection strategy, extended here to multi-view                   |
| NeuralRecon [12]                    | GRU-based multi-view fusion for real-time reconstruction           | Related multi-view fusion; this work substitutes attention for GRU fusion |
| DeepVideoMVS [13]                   | Convolutional-LSTM in place of 3D CNNs                             | Related lightweight-fusion motivation                                     |
| Attention-based aggregation [8, 10] | Learned per-view feature weighting                                 | Directly motivates the proposed attention module                          |
| This work                           | Geometry + learned odometry + attention-weighted multi-view fusion | Integrated end-to-end pipeline                                            |

### 3. Problem Formulation

The system is formulated as a mapping from an RGB image sequence to a volumetric scene representation. The input consists of an RGB frame sequence  $\{I_1, \dots, I_n\}$ , the camera intrinsic matrix  $K$ , and, for training, camera extrinsics  $[R|t]$  obtained from the visual odometry module or ground-truth capture metadata. Intermediate representations are: a reduced keyframe subset selected by the rotation/translation threshold of Section 5; per-keyframe 2D multi-scale features from the backbone and FPN; a sparse 3D voxel grid populated by projecting each voxel into every keyframe via  $P = K[R|t]$ ; and an attention-refined per-voxel feature after multi-view aggregation. The output is a per-voxel occupancy probability and truncated signed distance value, from which a surface mesh is extracted by marching cubes. Formally, the network learns a function  $f : \{I_1..I_n, K, \{R_i, t_i\}\} \rightarrow \{TSDF(v), occ(v)\}$  for each voxel  $v$  in the scene volume.

### 4. System Requirements and Feasibility

#### Hardware and Software

| Component | Requirement                             |
|-----------|-----------------------------------------|
| GPU       | NVIDIA A100, 40 GB                      |
| RAM       | Approximately 256 GB                    |
| CPU       | Intel i7, 4 cores, ~2.00 GHz base clock |

| Software / Library | Purpose                                           |
|--------------------|---------------------------------------------------|
| Python 3.8+        | Implementation language                           |
| NumPy              | Numerical / tensor processing                     |
| OpenCV             | Image processing, epipolar geometry, optical flow |
| PyTorch            | Deep learning, autograd, GPU training             |
| TorchVision        | Pretrained MNASNet backbone                       |
| scikit-image       | Marching cubes mesh extraction                    |
| Open3D             | Mesh / point-cloud visualization                  |
| TorchSparse        | Sparse 3D convolutions                            |
| Matplotlib         | Training-metric plots                             |

#### Functional and Non-Functional Requirements

Functionally, the system accepts an RGB video, selects keyframes by the ego-motion threshold, estimates per-keyframe camera pose, extracts and enhances 2D features, unprojects features into 3D, aggregates multi-view observations via attention, refines the resulting volume, and predicts TSDF/occupancy for mesh generation. Non-

functionally, the system targets usability (seamless video-to-mesh conversion), efficiency (expected mesh resolution within compute budget), and reliability (consistent output without security or correctness defects).

### Feasibility

The system is technically feasible, since all required geometric and deep-learning components are supported by existing libraries. It is operationally feasible on any system meeting the stated hardware/software baseline. Economically, training is computationally expensive, but inference is expected to require substantially less compute, keeping end-user cost low.

## 5. Dataset and Data Preparation

### Dataset Selection

Four candidate datasets were considered: ShapeNet (CAD-rendered multi-view objects), ScanNet (indoor scenes captured with an iPad LiDAR sensor), KITTI (outdoor stereo driving scenes), and Tanks and Temples (multi-view scenes of large structures). ScanNet was selected for its combination of high scene variation with a tractable, bounded structural complexity suited to indoor volumetric reconstruction, and because it directly provides RGB-D captures with LiDAR-derived depth usable for TSDF ground truth.

### ScanNet and Dataset Split

Each ScanNet scene provides 27 views captured with an iPad LiDAR sensor, from which ground-truth TSDF volumes are generated via TSDF fusion of the sensor depth maps; only the RGB stream and derived camera poses are used as network input, with the LiDAR-derived TSDF reserved as supervision.

| Split      | Number of Scenes |
|------------|------------------|
| Training   | 6,000            |
| Validation | 1,110            |
| Test       | 630              |

### Preprocessing and Keyframe Selection

Images are resized and normalized using ImageNet statistics, and camera intrinsics are rescaled to match. Per-scale projection matrices are computed as  $P = K[R|t]$ . Rather than processing every captured frame, the system selects a new keyframe only when the estimated ego-motion relative to the previous keyframe exceeds a rotation threshold of  $15^\circ$  or a translation threshold of 10 cm. This reduces redundant observations — consecutive high-frame-rate frames contribute little additional 3D structure — while retaining meaningful viewpoint changes needed for multi-view fusion.

## 6. Proposed Architecture

The pipeline shown in Fig. 1 comprises five stages beyond keyframe selection and visual odometry (Section 8): 2D feature extraction, feature enhancement, geometric unprojection, sparse 3D processing, and 3D feature refinement. Each is described below.

## 2D Feature Extraction: MNASNet

Each keyframe is passed through a 2D convolutional backbone to obtain dense image features. Because the downstream volume already carries substantial computational cost, a lightweight backbone is required. Among AlexNet, SqueezeNet, ResNet-18, MobileNetV2, MobileNetV3, MnasNet, and EfficientNet-B0, MnasNet was selected: its factorized hierarchical neural-architecture search allows different operations per block and mixes 3×3 and 5×5 convolutions, consistently outperforming MobileNetV3 at comparable depth multipliers while remaining lighter than ResNet-18 or EfficientNet-B0. The backbone is initialized with ImageNet-pretrained weights and frozen during training, since its convolutional layers already encode useful low-level image structure.



Figure 7: MnasNet backbone architecture used for 2D feature extraction.

### Feature Enhancement: Feature Pyramid Network

Early backbone layers yield higher-resolution but less semantically detailed features, while later layers yield lower-resolution but more semantically detailed features. A Feature Pyramid Network (FPN) combines both: features at backbone scales  $\times 8$  and  $\times 16$  are projected to a common dimension via linear layers, the coarser features are upsampled and added to the finer features through lateral connections, and the fused result is passed through a convolutional layer for spatial refinement, yielding a feature map that is simultaneously high-resolution and high-semantic.

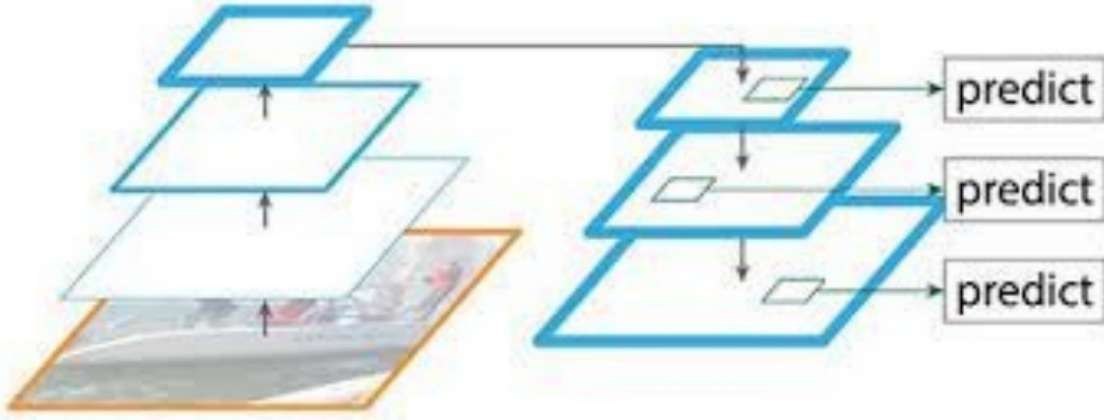


Figure 8: Feature Pyramid Network (FPN) combining multi-scale MnasNet features via top-down lateral connections.

### Geometric 2D-to-3D Unprojection

Geometric unprojection establishes the correspondence between image-space features and spatial locations in the reconstructed volume. A volume of the world is discretized into voxels; for each voxel, the projection matrix  $P_i = K[R_i | t_i]$  of every keyframe determines its projected 2D coordinate in that view. If the projected coordinate falls within the valid image region, the corresponding image feature is a valid observation of that voxel; if it falls outside the image bounds, the observation is masked out. Formally, given a voxel  $X$ ,  $[x_1 \dots x_n] = [P_1 \dots P_n]X$ , and the sampled features at  $[x_1 \dots x_n]$  become the set of valid per-view observations available for that voxel. Where multiple views observe the same voxel, a baseline (non-attention) implementation simply sums and averages the available features using a validity mask to avoid division by zero; Section 7 replaces this step with learned attention weighting.

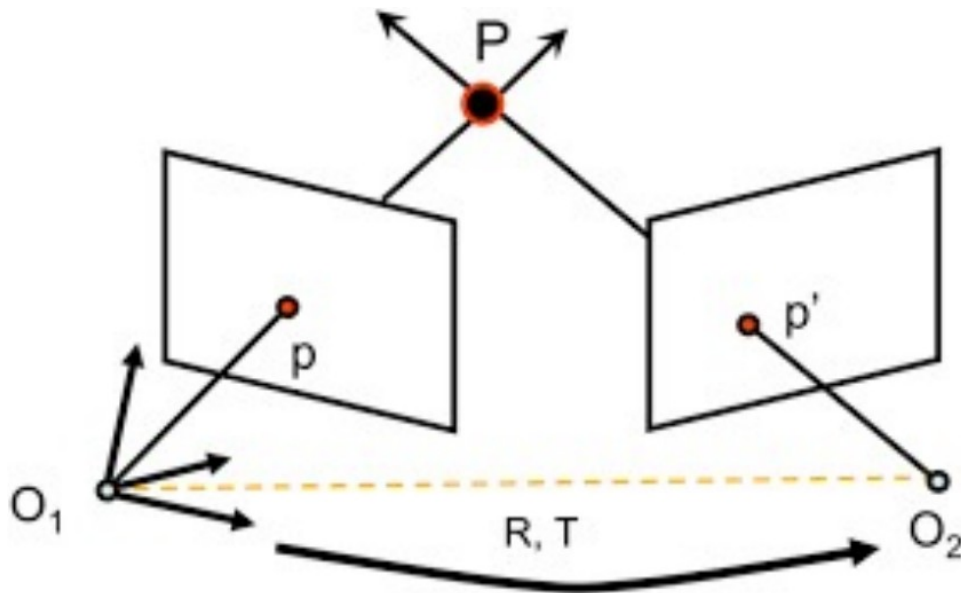


Figure 9: Geometric unprojection: each voxel accumulates image features from every keyframe whose projection ray passes through it.

### Sparse 3D Processing

Most voxels in an indoor scene volume are unoccupied. Performing dense 3D convolutions over the full grid wastes computation on empty space, so the feature volume is instead stored and processed as a sparse tensor using TorchSparse, which is built on Google's sparse hash library and a CUDA-native implementation reported to run roughly 3× faster than the alternative Minkowski Engine. This allows the refinement network to operate only on occupied or near-surface regions of the volume.

### 3D Feature Refinement: Residual U-Net

The unprojected (and attention-refined) volume is passed through a U-Net-style network for spatial refinement before TSDF/occupancy prediction. A vanilla U-Net was first tried and found to overfit given the available training data, so it was replaced with a Residual U-Net, adding residual connections at each stage. The contraction path (downsampler) records scene context through convolution and max-pooling; the symmetric expansion path (upsampler) recovers precise localization via transposed convolutions and concatenation with the corresponding downsampler features.

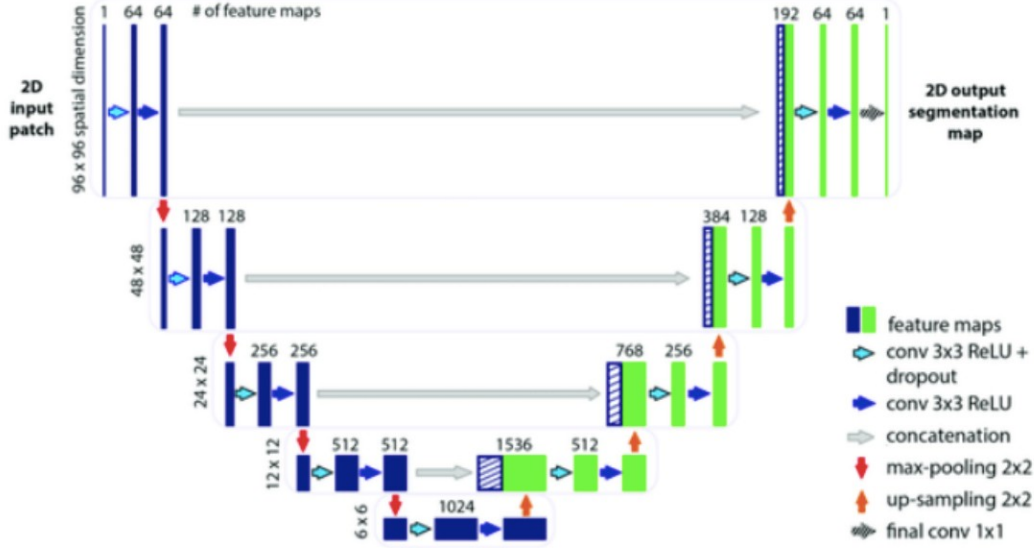


Figure 10: Residual U-Net architecture used for volumetric feature refinement.

## 7. Attention-Based Multi-View Aggregation

### Motivation

Multiple keyframe observations frequently project onto the same voxel, but these observations need not carry equally useful information — some views may see the surface point head-on and in sharp focus, while others observe it at a glancing angle or partially occluded. Direct additive accumulation, as used in the SurfaceNet/NeuralRecon-style baseline, treats every valid observation identically and does not explicitly learn which views are more informative for a given voxel. The proposed attention mechanism introduces learned, per-observation weighting during multi-view fusion in place of this uniform accumulation.

### Inputs, Query, and Key

The module operates on two feature sets already available from earlier stages: the projected per-pixel image features obtained via the unprojection procedure of Section 6.3, and the current point (voxel) feature carried forward from the previous refinement step. The point feature is passed through a linear layer to obtain a query vector, and the pixel features are passed through a separate linear layer to obtain key vectors — one key per valid observation of that voxel.

### Attention Score and Weighted Aggregation

A scaled dot product between the query and each key,  $\text{query} \cdot \text{key}^T / D$  ( $D$  the feature dimension), produces a raw attention score per observation. Scores corresponding to pixel rays that do not pass through the voxel are masked to a large negative value so that they vanish after normalization. A softmax over the remaining scores yields per-view attention weights, and the pixel (key) features are combined via this weighted sum rather than a uniform average. The result is added to a residual linear projection of the original point feature and passed through two 3D convolutional layers before continuing to the refinement network of Section 6.5.

#### 4.4.6. Novel architecture

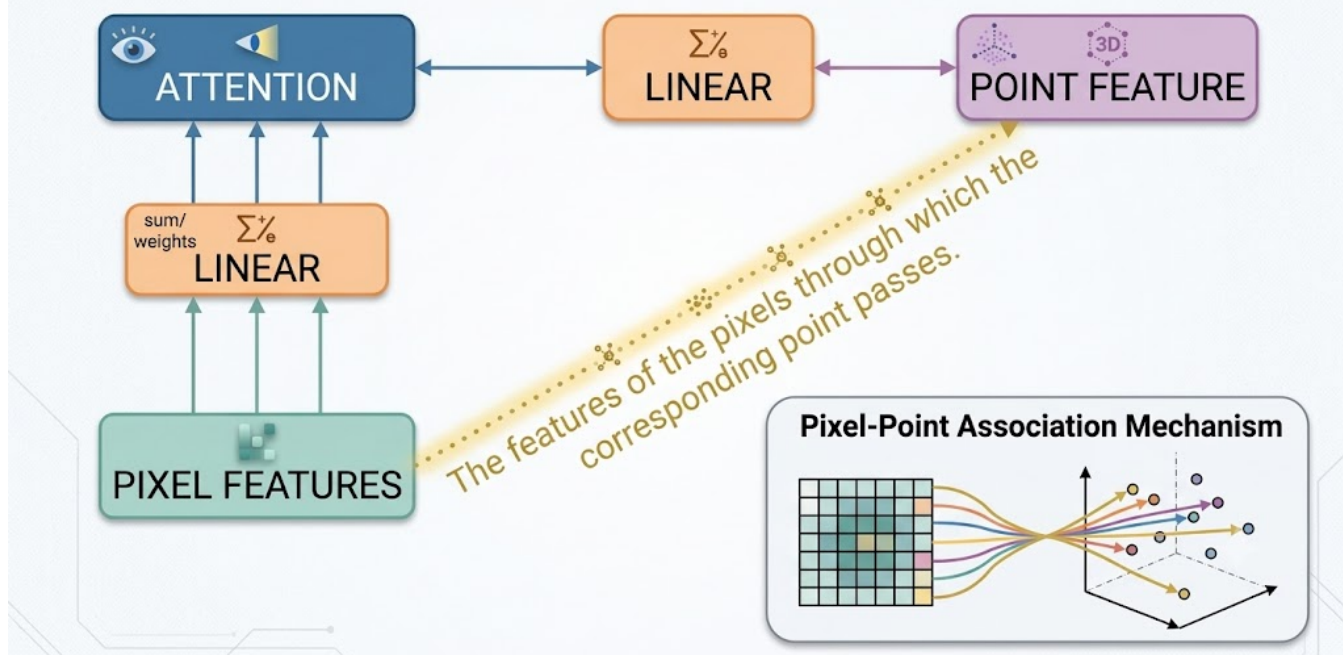


Figure 11: Attention-based accumulation replaces additive voxel-feature accumulation with a learned, weighted combination.

#### Integration into the Pipeline

The attention module sits directly between the unprojection stage and the 3D refinement network: MnasNet → FPN → Unprojection → Attention-Weighted Aggregation → Residual U-Net → TSDF/Occupancy Head. It does not alter any upstream stage; it only changes how multi-view observations are combined once they arrive at a shared voxel.

#### Why This Is Expected to Help

By learning per-observation relevance rather than fixing it to a uniform average, the network can, in principle, down-weight low-quality or misleading observations (e.g., grazing-angle or occluded views) and up-weight more informative ones, at each voxel independently. We do not claim this improvement is universal or guaranteed for every scene configuration; Section 10 reports the empirically observed effect on both training dynamics and final reconstruction accuracy for the evaluated dataset.

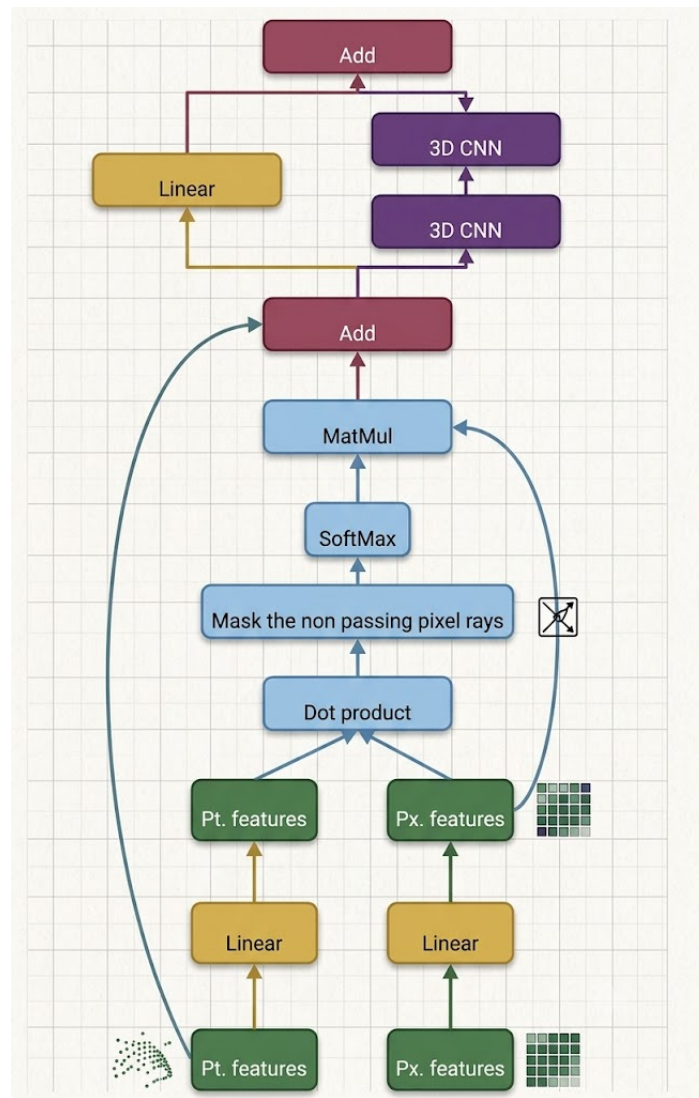


Figure 12: Detailed attention block: query/key linear projections, masked scaled dot-product attention, softmax normalization, and residual fusion with 3D convolutions.

## 8. Visual Odometry

### Purpose

Every stage of the reconstruction pipeline depends on knowing each keyframe's camera pose relative to a common world frame. The visual odometry module estimates this pose — six degrees of freedom comprising translation (horizontal, vertical, depth) and rotation (pitch, yaw, roll) — for each pair of consecutive frames, combining classical geometry with a learned refinement stage (Fig. 13).

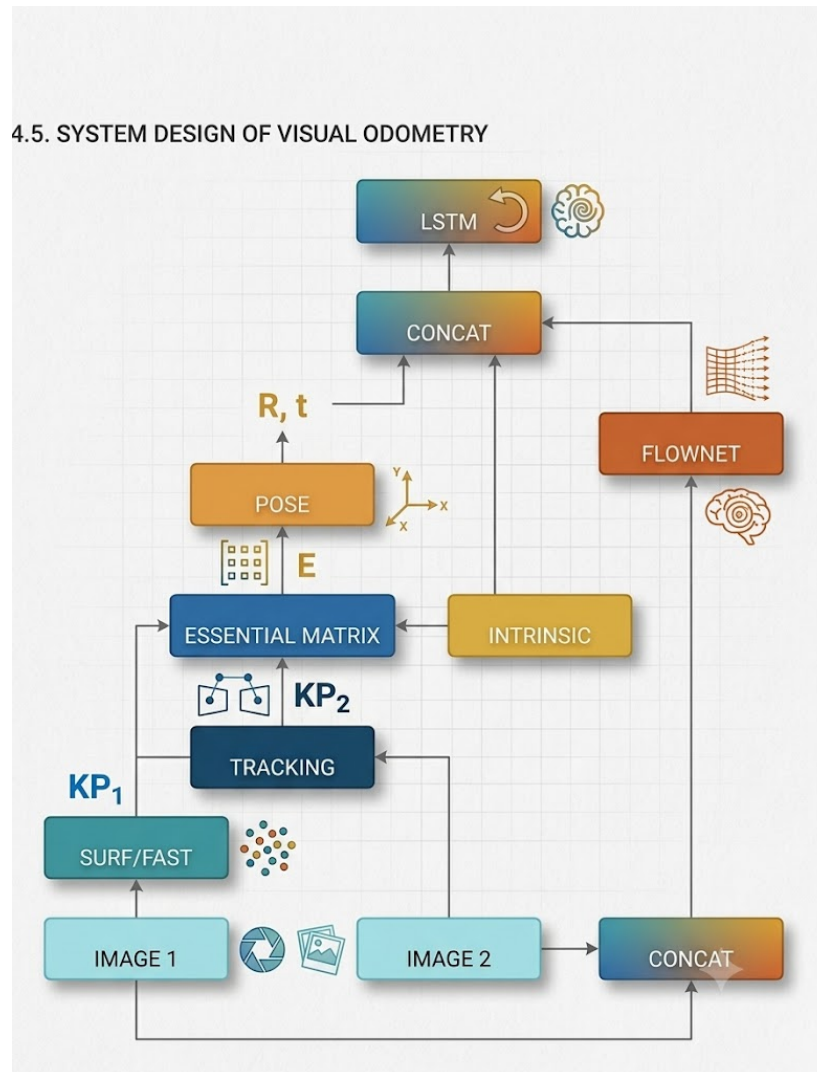


Figure 13: Hybrid geometric and deep-learning visual odometry pipeline.

### Geometric Estimation

Rather than re-matching descriptors between consecutive frames, keypoints are tracked using optical flow, which estimates the coordinates of a keypoint in the current frame given its location in the reference frame. Pose is then recovered via the essential matrix  $E = K^T F K$ , satisfying  $x'^T K'^{-T} E K^{-1} x = 0$ , rather than the fundamental matrix directly. Because camera intrinsics are supplied explicitly, only five point correspondences are required to solve for  $E$  (Nister's five-point algorithm), improving accuracy over the fundamental-matrix formulation used in the classical baseline of Section 2.1.

## Learned Refinement: FlowNet and LSTM

FlowNet predicts the optical flow between a concatenated image pair as a dense vector field. This flow vector is concatenated with the geometric rotation/translation estimate and with the camera intrinsics, giving the model access to the full camera configuration. The combined vectors for all consecutive frame pairs in a sequence are fed to an LSTM, which predicts the current-frame projection matrix while retaining state from previous timesteps — effectively refining the geometric estimate using a learned temporal model rather than replacing it outright.

## Scale Ambiguity

A strictly monocular visual odometry system cannot recover absolute scale from image data alone; scale can only be recovered independently by a stereo configuration. The scale-corrected geometric result reported in Section 10 uses scale computed from ground truth, which is unavailable at real-time inference. This is an important practical limitation: the combined geometric-and-deep-learning method reported in Section 10 achieves its accuracy without scale correction, but absolute real-world scale for a purely monocular deployment remains an open problem for this pipeline.

# 9. Implementation and Training

## Data Loader

A custom data loader retrieves mini-batches dynamically rather than loading the full dataset into memory, and jointly handles the multiple input modalities required: RGB images, camera intrinsics, camera extrinsics (from visual odometry), the TSDF ground truth (from TSDF fusion), and the scene origin (initial point of all camera frusta).

## Model, Optimizer, and Loss

The model is implemented via subclassing rather than the Sequential or Functional APIs, since its multi-modal, multi-stage structure exceeds what those simpler APIs support cleanly. The Adam optimizer is used with a learning rate of 0.001. Voxel occupancy, a Boolean quantity, is supervised with binary cross-entropy loss; TSDF value regression, a real value bounded to  $[0, 1]$ , is supervised with an L1 loss between predicted and ground-truth values. Regularization is provided by dropout and data augmentation, introduced together with the attention module (Section 10.2).

## Core Routines and Pipeline Summary (Pseudocode)

The four routines below summarize, in order: world-grid generation, the baseline (additive) unprojection path, the attention-weighted aggregation that replaces it, and the overall per-sequence pipeline.

Listing 1: World-grid generation for a given voxel scale.

```
function BUILD_WORLD_GRID(dims, interval, origin):
  x_range <- range(0, dims[0] // interval)
  y_range <- range(0, dims[1] // interval)
  z_range <- range(0, dims[2] // interval)
  grid <- meshgrid(x_range, y_range, z_range) # unit-length voxel grid
  world_grid <- stack(grid)*interval + origin # scale + translate to scene origin
  return world_grid # (N,3) voxel coords in world space
```

Listing 2: Unprojection with validity masking (baseline additive path).

```

function SAMPLE_AND_ACCUMULATE(features, voxel_coords, proj_matrices):
    pixel_coords <- PROJECT(proj_matrices, voxel_coords) # [P_1 ... P_n] X
    mask <- IN_BOUNDS(pixel_coords) # valid = ray falls in image
    sampled <- GRID_SAMPLE(features, pixel_coords) # bilinear sample per view
    sampled[NOT mask] <- 0.0 # zero-out invalid observations
    voxel_features <- SUM(sampled, over_views)
    voxel_mask <- SUM(mask, over_views)
    voxel_mask[voxel_mask == 0] <- 1 # avoid divide-by-zero
    voxel_features <- voxel_features / voxel_mask # additive-average baseline
    return voxel_features

```

**Listing 3: Attention-weighted aggregation, replacing the additive-average step above.**

```

function ATTENTION_AGGREGATE(point_feature, pixel_features, mask, D):
    query <- LINEAR_Q(point_feature) # W_query
    key <- LINEAR_K(pixel_features) # W_key
    scores <- MATMUL(query, TRANSPOSE(key)) / sqrt(D)
    scores[NOT mask] <- -10000.0 # mask rays that miss the voxel
    weights <- SOFTMAX(scores, axis=views)
    fused <- MATMUL(weights, key) # weighted sum over valid views
    output <- fused + LINEAR_RESIDUAL(point_feature)
    output <- CONV3D(CONV3D(output))
    return output

```

**Listing 4: Per-sequence pipeline summary.**

```

for each input frame: # Sec. 5
    estimate camera motion via visual odometry # Sec. 8
    if motion exceeds rotation/translation threshold: select frame as keyframe
for each keyframe:
    extract 2D features (MnasNet) | enhance multi-scale features (FPN) # Sec. 6
for each voxel in scene volume:
    project voxel into every keyframe using  $P = K[R|t]$  # Sec. 6.3
    collect valid per-view image features
    aggregate features via attention-weighted sum # Sec. 7
    refine aggregated volume (Residual U-Net, sparse 3D convs) # Sec. 6.5
    predict occupancy and TSDF
extract surface mesh via marching cubes

```

## 10. Experimental Evaluation and Results

The evaluation addresses three questions: (i) does attention-based aggregation reduce TSDF reconstruction loss relative to additive fusion; (ii) does combining learned and geometric information improve visual odometry accuracy; and (iii) how does the predicted geometry compare qualitatively with LiDAR-derived ground truth?

### Effect of Attention on TSDF Reconstruction

Before adding attention, validation loss plateaus after a limited number of training iterations (Fig. 14). After adding attention, together with dropout and data augmentation, validation loss continues tracking training loss over a longer horizon despite training being harder, yielding lower final test loss (Fig. 15).

6.1. TESTING OF NOVEL ARCHITECTURE

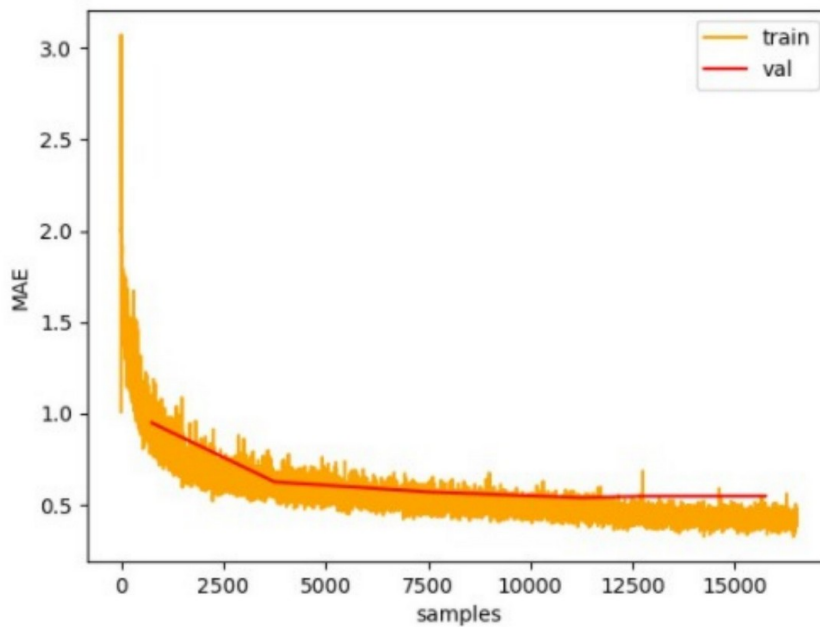


Figure 14: Training/validation MAE before adding attention — validation loss stagnates early.

| Configuration                      | TSDF L1 Loss |
|------------------------------------|--------------|
| Before attention (additive fusion) | 1.23         |
| After attention (weighted fusion)  | 0.93         |

The 24% relative reduction in L1 loss indicates that learned, per-view weighting of multi-view observations measurably improves TSDF regression accuracy on ScanNet relative to uniform additive accumulation, consistent with the motivation of Section 7.1.

### Voxel Occupancy Classification

The coarse voxel layer, which establishes basic scene structure, is dominated by empty (vacuum) voxels. At a zero decision threshold: true negative 0.9200, false negative 0.000353, false positive 0.04826, true positive 0.03138. Because a false negative (an occupied voxel misclassified as empty) destroys real scene structure, while a false positive can subsequently be corrected during TSDF regression, the decision threshold was shifted toward the

negative class, reducing the coarse-layer false-negative rate to roughly  $4 \times 10^{-6}$  (true negative 0.8621, false positive 0.1058, true positive 0.03209). The fine voxel layer, capturing finer intra-voxel detail, is more class-balanced (true negative 0.4837, true positive 0.4519 at zero threshold); the same threshold shift reduced its false-negative rate from 0.02923 to 0.01996.

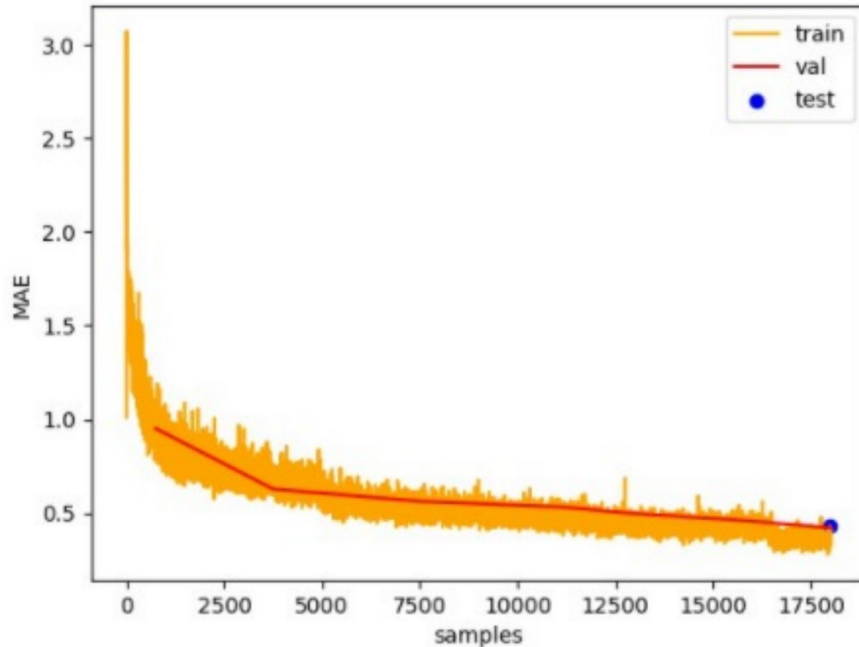


Figure 15: Training/validation/test MAE after adding attention — validation loss keeps pace with training through convergence.

## Visual Odometry Accuracy

| Method                                             | End-Point Error |
|----------------------------------------------------|-----------------|
| Geometry only, without scale correction            | 0.2151          |
| Geometry, with ground-truth scale correction       | 0.1478          |
| Deep learning + geometry, without scale correction | 0.0919          |

The combined geometric-and-deep-learning method achieves the lowest end-point error of the three configurations despite operating without scale correction, which is only computable from ground truth and therefore unavailable to a real-time monocular system (Section 8.4). This suggests the learned refinement stage compensates for some of the accuracy that scale correction would otherwise provide, though it does not resolve the underlying scale-ambiguity limitation.

## Qualitative Reconstruction Results

Meshes were extracted from the predicted TSDF volume via marching cubes and compared, from matching viewpoints, against the LiDAR-fused ground-truth mesh for held-out ScanNet test scenes. The attention-enabled predicted meshes closely match ground truth in overall structure — walls, furniture silhouettes, and large planar surfaces — though fine surface detail and thin structures remain somewhat less precise than the LiDAR-derived reference.

Across all ten evaluated test scenes, the attention-enabled predicted meshes consistently reproduce the large-scale structure of the LiDAR-fused ground truth, including walls, furniture silhouettes, and dominant planar surfaces, with the same tendency toward softer fine-surface and thin-structure detail noted above.

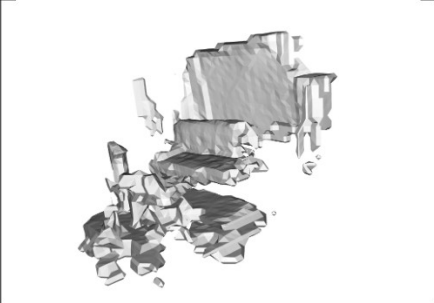
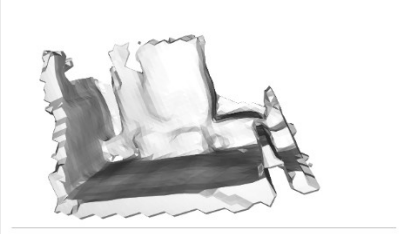
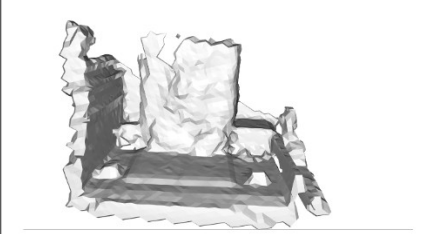
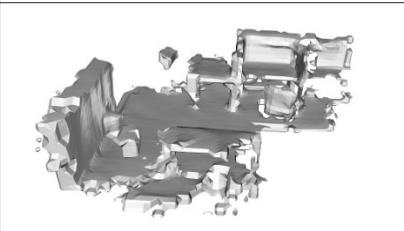
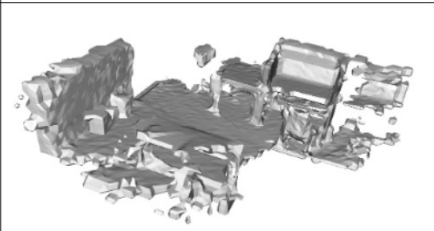
| <div>Scenario</div> | <div>Predicted Mesh<br/>(Predicted by the model)</div>                             | <div>Fused Mesh<br/>(Ground Truth)</div>                                            |
|---------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1                   |   |   |
| 2                   |   |   |
| 3                   |  |  |

Figure 16: Predicted mesh vs. LiDAR-fused ground-truth mesh, representative test scenes 1–3.

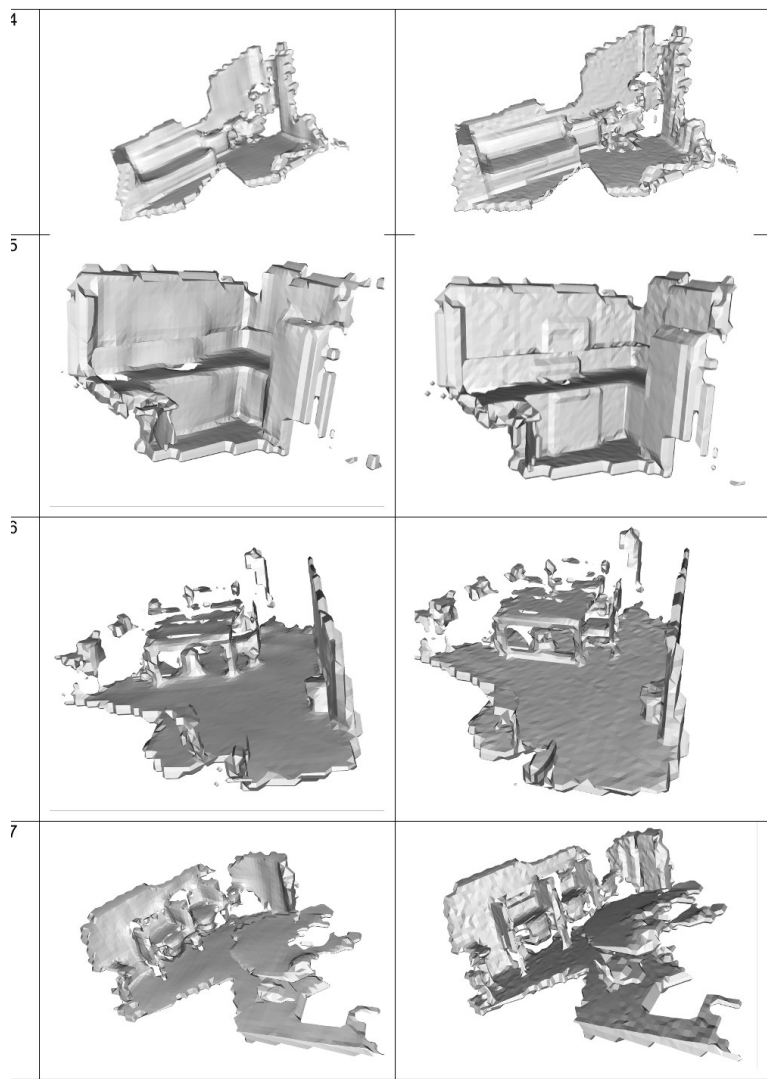


Figure 17: Predicted mesh vs. LiDAR-fused ground-truth mesh, representative test scenes 4–7.

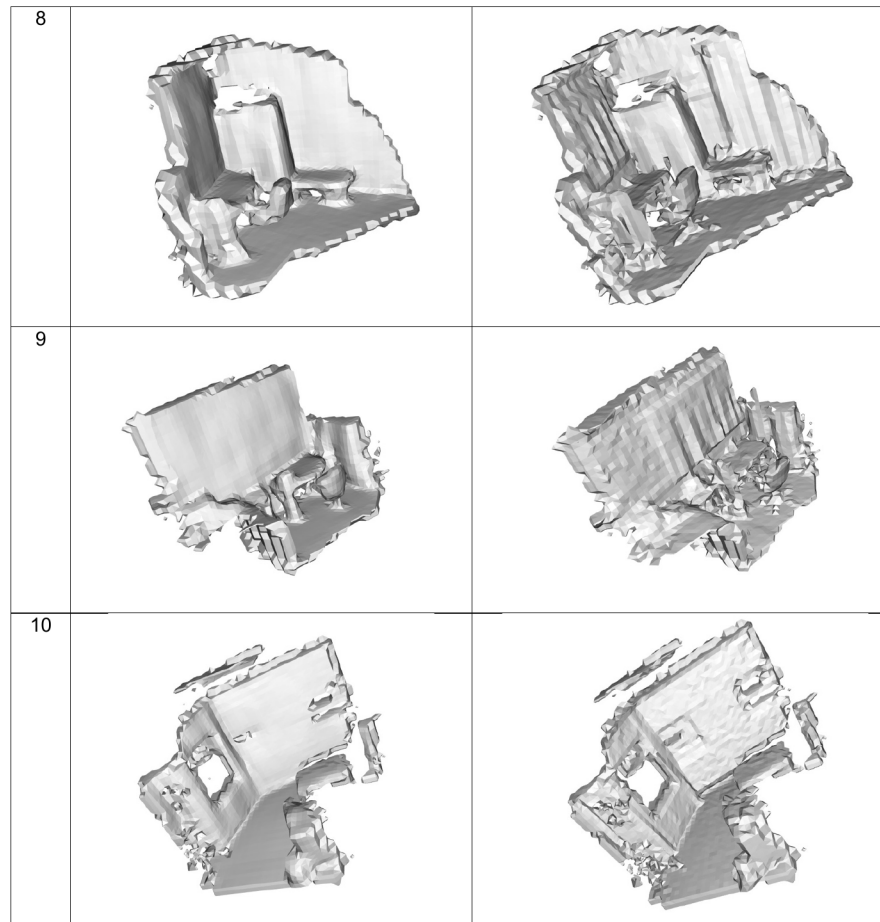


Figure 18: Predicted mesh vs. LiDAR-fused ground-truth mesh, representative test scenes 8–10, continuing the qualitative comparison of Fig. 12.

## 11. Discussion, Limitations, Future Work, and Conclusion

### Discussion

The reported TSDF L1 loss reduction from 1.23 to 0.93 indicates that learned attention weighting provides a real, measurable benefit over uniform multi-view accumulation on ScanNet, though the underlying refinement network, backbone, and training data are unchanged between the two configurations — the improvement is attributable specifically to how multi-view observations are combined at each voxel. Similarly, the visual-odometry results show that a learned refinement stage can substantially reduce end-point error relative to a purely geometric estimate, even without access to scale information. Together, these results support the hypothesis that combining classical geometric structure with learned, data-driven weighting is more effective than either approach used in isolation for this task.

### Limitations

Three limitations constrain the system's current applicability. First, dense-adjacent 3D convolutions in the refinement network dominate inference latency, and despite sparse-tensor processing, the pipeline is not currently suitable for real-time operation. Second, monocular visual odometry cannot recover absolute scale independently; the reported odometry results depend on ground-truth scale correction for one configuration and omit it for the best-performing configuration, meaning real-world absolute scale remains unresolved for a purely monocular deployment. Third, evaluation is limited to the ScanNet indoor-scene distribution; generalization to outdoor, large-scale, or dynamic scenes has not been evaluated and cannot be assumed.

### Future Work

Because attention allows each voxel to retain the most relevant features among its contributing views, a natural extension is applying the same mechanism to 3D semantic segmentation. Since 3D convolutions dominate inference latency, a further planned enhancement is reimplementing the attention module in C++ with Python bindings to reduce runtime. Incorporating a stereo or multi-camera configuration to resolve scale independently of ground truth would also address the odometry module's principal limitation.

### Conclusion

This work presented a sensor-free, RGB-only multi-view 3D reconstruction pipeline that combines geometric visual odometry, lightweight 2D feature extraction, geometric unprojection, and a novel attention-based multi-view feature-aggregation mechanism. On ScanNet, the attention mechanism reduced TSDF regression loss from 1.23 to 0.93, and a hybrid geometric-and-deep-learning odometry estimator outperformed either component used alone. While real-time deployment remains unmet due to 3D convolution cost and monocular scale ambiguity, the results support attention-weighted multi-view fusion as a promising direction for sensor-free 3D scene reconstruction.

## References

[1] D. G. Lowe, “Distinctive Image Features from Scale-Invariant Keypoints,” *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.

- [2] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge, U.K.: Cambridge University Press, 2004.
- [3] S. Wang, R. Clark, H. Wen, and N. Trigoni, “DeepVO: Towards End-to-End Visual Odometry with Deep Recurrent Convolutional Neural Networks,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2017, pp. 2043–2050.
- [4] L. Mescheder, M. Oechsle, M. Niemeyer, S. Nowozin, and A. Geiger, “Occupancy Networks: Learning 3D Reconstruction in Function Space,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 4455–4465.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [6] M. Tan et al., “MnasNet: Platform-Aware Neural Architecture Search for Mobile,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 2815–2823.
- [7] Q. Xu, W. Wang, D. Ceylan, R. Mech, and U. Neumann, “DISN: Deep Implicit Surface Network for High-Quality Single-View 3D Reconstruction,” in *Proc. NeurIPS*, 2019.
- [8] A. Salvi, N. Gavenski, E. Pooch, F. Tasoniero, and R. Barros, “Attention-Based 3D Object Reconstruction from a Single Image,” in *Proc. Int. Joint Conf. Neural Networks (IJCNN)*, 2020, pp. 1–8.
- [9] A. Thai, S. Stojanov, V. Upadhyay, and J. M. Rehg, “3D Reconstruction of Novel Object Shapes from Single Images,” in *Proc. Int. Conf. 3D Vision (3DV)*, 2021, pp. 85–95.
- [10] B. Yang, S. Wang, A. Markham, and N. Trigoni, “Robust Attentional Aggregation of Deep Feature Sets for Multi-View 3D Reconstruction,” *Int. J. Comput. Vision*, vol. 128, pp. 53–73, 2020.
- [11] M. Ji, J. Gall, H. Zheng, Y. Liu, and L. Fang, “SurfaceNet: An End-to-End 3D Neural Network for Multiview Stereopsis,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2326–2334.
- [12] J. Sun, Y. Xie, L. Chen, X. Zhou, and H. Bao, “NeuralRecon: Real-Time Coherent 3D Reconstruction from Monocular Video,” in *Proc. CVPR*, 2021.
- [13] A. Düzceker, S. Galliani, C. Vogel, P. Speciale, M. Dusmanu, and M. Pollefeys, “DeepVideoMVS: Multi-View Stereo on Video with Recurrent Spatio-Temporal Fusion,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 15319–15328.
- [14] A. Dai, A. X. Chang, M. Savva, M. Halber, T. Funkhouser, and M. Nießner, “ScanNet: Richly-Annotated 3D Reconstructions of Indoor Scenes,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2432–2443.
- [15] H. Tang, Z. Liu, X. Li, Y. Lin, and S. Han, “TorchSparse: Efficient Point Cloud Inference Engine,” in *Proc. Conf. Machine Learning and Systems (MLSys)*, 2022.